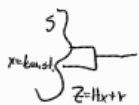


1) Meritve konst. kol.

$\{z_i\}_n$



$$\frac{dP}{dr} \sim N(0, \sigma)$$

$n \rightarrow \infty$

histogram $\rightarrow \mu, \sigma$



n omejen

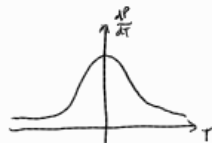
vzorčna statistika

$$\begin{aligned} \bar{z} &= \frac{1}{n} \sum z_i && \text{povprečje vzorca} \\ s^2 &= \frac{1}{n-1} \sum (z_i - \bar{z})^2 && \text{vzorčna varianca} \end{aligned} \quad \left. \vphantom{\sum} \right\} \text{cenilski/vzorčni statistiki}$$

konstruiramo statistiko T

$$T = \frac{(\bar{z} - \mu)}{\frac{s}{\sqrt{n}}}$$

$$\text{Studentov zakon: } \frac{dP}{dT} = \frac{1}{\Gamma(n/2) \beta^{n/2}} \left(1 + \frac{T^2}{n}\right)^{-n/2}$$



$$\frac{dP}{dT} \neq f(\mu)$$

zvečana oblika

$$T \in [t_c, t_2]$$

$$t_c \leq \frac{(\bar{z} - \mu)\sqrt{n}}{s} \leq t_2$$

$$\int_{t_c}^{t_2} \frac{dP}{dT} dT = 1 - \alpha$$

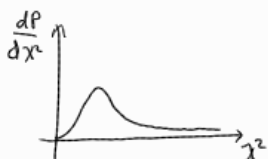
$$\mu \in \left[\bar{z} - \frac{t_2 s}{\sqrt{n}}, \bar{z} - \frac{t_c s}{\sqrt{n}} \right] \quad \text{če } \mu \text{ pade izven intervala, ga lahko}$$

na stopnji tveganja α zavrzemo

ocenjena disperzija

Porazdelitev χ^2

$$\chi^2 = (n-1) \frac{s^2}{\sigma^2} \quad \frac{dP}{d\chi^2} = \frac{1}{2^{n/2}} \frac{1}{\Gamma(n/2)} (\chi^2)^{n/2-1} e^{-\chi^2/2}$$



$$\frac{dP}{d\chi^2} \neq f(s, \sigma)$$

Interval zaupanja

$$\chi^2 \in [\chi^2_c, \chi^2_2] \text{ z } (1-\alpha)$$

na tveganje:

$$\chi^2_s = \frac{(n-1)s^2}{\sigma^2}$$

$$\sigma \in [\sigma_c, \sigma_2] \text{ na stopnji zaupanja } (1-\alpha)$$

Def: Vsota kvadratov n neodvisnih po $N(0,1)$ porazdeljenih spremenljivk je porazdeljena po univerzalnem zakonu χ^2 .

$$z_i \sim N(s, \sigma) \quad n \text{ meritev}$$

$$\frac{(z_i - \bar{z})}{\sigma} \sim N(0,1) \quad n-1 \text{ neodvisnih}$$

$$\rightarrow \sum \left(\frac{(z_i - \bar{z})}{\sigma} \right)^2 = \frac{s^2}{\sigma^2} (n-1)$$

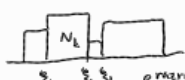
$$\text{Def: } \begin{cases} X \text{ por. po } N(0,1); Y \sim \chi^2(n) \\ \rightarrow T = \frac{X}{\sqrt{Y/n}} \sim S(n) \end{cases}$$

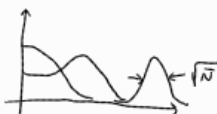
studentova porazd.

$$T = \frac{\bar{z} - \mu}{\frac{s}{\sqrt{n}}} \sqrt{n} = \frac{\bar{z} - \mu}{\frac{s}{\sqrt{n}}} \frac{1}{\sqrt{\frac{s^2}{\sigma^2} \frac{n-1}{n}}} = \frac{X}{\sqrt{Y/n}} \sim S(n-1)$$

$$X \sim N(0,1) \quad Y \sim \chi^2(n-1)$$

Oblikovni testi

$\frac{dP}{d\theta}$

 N - vseh izmerkov
 Np_k - pričakovano št. izmerkov v k-tom razredu
 N_k - št. izmerkov v k-tom razredu, $N_k \geq 5$
 $P_k = \int_{g_{k-1}}^{g_k} \frac{dP}{d\theta} d\theta$
 ocenjeno $\frac{Np_k - N_k}{\sqrt{Np_k}}$

$\frac{dP}{dN} = \frac{\bar{N}^N e^{-\bar{N}}}{N!} \xrightarrow{N \rightarrow \infty} N(\bar{N}, \sqrt{\bar{N}}) \rightarrow \frac{N_k - Np_k}{\sqrt{Np_k}} \sim N(0,1)$

 $\chi^2 = \sum_{k=1}^g \frac{(N_k - Np_k)^2}{Np_k} \sim \chi^2(g-1)$
 Pearsonov χ^2 test

Primer



$$Np_k = (5, 4, 8, 6, 7) \frac{637}{26} = (123,5; 98; 196; 147; 73,5)$$

$k=1$ $k=2$ $k=3$ $k=4$ $k=5$

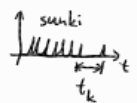
$$\chi^2 = \sum \frac{(N_k - Np_k)^2}{Np_k} = \dots = 9,27\%$$

$$g-1=4$$

$$P(\chi^2 > \chi^2_c) = 0,05 \rightarrow \chi^2_c = 9,49$$

→ na stopnji tvegaja 5% ne moremo zavrniti hipoteze

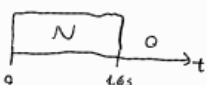
RA razpad


 $e^{-t/\tau}$ da ne razpade
 $dt \frac{1}{\tau}$ da razpade
 $\frac{dP}{dt} = \frac{1}{\tau} e^{-t/\tau}$ preverjamo
 $\frac{dP}{dt} = ?$

100 parov ($N=100$)

$t_{max} = 1,6s$

Hipoteza: $\frac{dP}{dt} = \frac{1}{\tau} e^{-t/\tau}$; $\tau = 1s$



$$P_0 = \int_0^{1,6} \frac{1}{\tau} e^{-t/\tau} dt = -e^{-t/\tau} \Big|_0^{1,6} = 1 - e^{-1,6} = 0,8$$

$$P_1 = 1 - P_0 = 0,2$$

$$\chi^2 = \frac{(100-80)^2}{80} + \frac{(0-20)^2}{20} = 25$$

$$P(\chi^2 > \chi^2_c) = 5\% \rightarrow \chi^2_c = 5,99$$

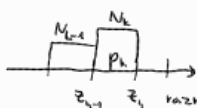
Na stopnji tvegaja 5% lahko zavrnemo hipotezo

Funkcije zanesljivosti in Fisherjev test

$$\frac{dP}{dz}(q_1, \dots, q_m)$$

↳ m možn. parametrov

določanje τ : (optimalno) iz izmerkov



razredi
 - uresni preizkusni prost.
 - se ne prekrivajo

$$P_k = \int_{z_{k-1}}^{z_k} \frac{dP}{dz}(q_1, \dots, q_n) dz = P_k(q_1, \dots, q_n)$$

funkcija zanesljivosti

$$L^* = \prod_{k=1}^N P_k(q_1, \dots, q_n)$$

za optimalne parametre ima L^* ekstrem $\rightarrow \frac{\partial L^*}{\partial q_i} = 0 \rightarrow q_i^*$ (optimalni q_i)

$$p_k(q_i^*) \rightarrow \text{Pearson } \chi^2$$

$$\chi^2 = \sum \frac{(N_k - N p_k^*)}{N p_k^*} = \chi^2(p-1-m)$$

$\hookrightarrow p-1-m$ prostih parametrov

Poenostavljena funkcija zanesljivosti

$$z_i - z_{i-1} \rightarrow 0 \quad (\text{vedno več razredov})$$

$$\rightarrow N_k = \sum_{i=1}^{\infty} \dots$$

$$L = \prod_k \left(\int_{z_{k-1}}^{z_k} \frac{dF}{dz} dz \right)^{N_k} = \prod_{\text{pobli } z_{k-1}}^{z_k} \left(\int_{z_{k-1}}^{z_k} \frac{dF}{dz} dz \right)^{N_k}$$

$$\frac{\partial L}{\partial q_i} = 0 \rightarrow q_i^* \quad \frac{\partial L}{\partial q_i} = 0 \Leftrightarrow \frac{\partial \ln L}{\partial q_i} = 0$$

Primer: RA razpad

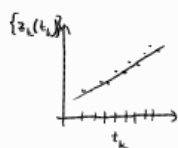
$$\{t_k\}; \quad \frac{dp}{dt} = \frac{1}{\tau} e^{-t/\tau}, \quad \tau = ?$$

$$L^* = \prod \frac{1}{\tau} e^{-t_k/\tau} = \left(\frac{1}{\tau}\right)^N \prod e^{-t_k/\tau}$$

$$\ln L^* = -N \ln \tau - \sum t_k/\tau = -N \ln \tau - \frac{1}{\tau} \sum t_k$$

$$\frac{\partial \ln L^*}{\partial \tau} = -\frac{N}{\tau} + \frac{1}{\tau^2} \sum t_k = 0$$

$$\tau = \frac{1}{N} \sum t_k = \bar{t}$$



fit funkcija $f(t_k)$

$(z_k - f(t_k)) \sim \text{Normalno porazdel. 3\sigma}$

$$L = \prod \exp\left[-\frac{1}{2} \frac{(z_k - f(t_k))^2}{\sigma_k^2}\right]$$

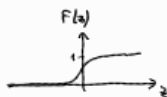
$$\ln L = -\frac{1}{2} \sum \frac{(z_k - f(t_k))^2}{\sigma_k^2}$$

$$\frac{\partial^2 \ln L}{\partial q_i \partial q_j} = R_{ij}^{-1}$$

Test Kolmogorova

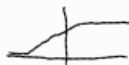
razredi $\{z_i\}$

$$F(z) = \int_{-\infty}^z \frac{dF}{dz} dz$$



kumulativni porazdel.

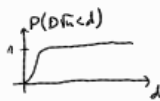
$$f(z) = \frac{\# \text{ izmerkov} < z}{n}$$



eksperimentalni kum. porazdel.

$$D = \sup |f(z) - F(z)|$$

$$P(D \leq d) = \sum_{k=0}^{\infty} (-1)^k e^{-k^2 d^2}$$



$D \leq d_c \rightarrow$ na stopnji α trdnost α lahko zavrnemo hipotezo

$D \leq d_c \rightarrow$ na st. α ne moremo zavrniti hipoteze

Metoda najin. kvadratov

$$\begin{bmatrix} z_1 \\ z_2 \\ \vdots \end{bmatrix} = Hx + v \rightarrow Hx = \begin{bmatrix} x_0 f_0(t_1) + x_1 f_1(t_1) + \dots + x_m f_m(t_1) \\ \vdots \\ x_0 f_0(t_n) + \dots + x_m f_m(t_n) \end{bmatrix}$$

$$H^T z = \begin{bmatrix} f_0(t_1) & f_1(t_1) & \dots & f_m(t_1) \\ \vdots & \vdots & \dots & \vdots \\ f_0(t_n) & f_1(t_n) & \dots & f_m(t_n) \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_m \end{bmatrix}$$

Kalman:

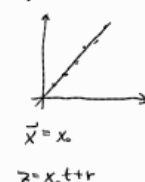
$$\hat{x} = \frac{1}{\sigma^2} P H^T R^{-1} (z - H \hat{x})$$

$$P^{-1} = \frac{1}{\sigma^2} P^{-1} + H^T R^{-1} H \quad R = \sigma^2 I$$

$$P^{-1} = \frac{1}{\sigma^2} H^T H \rightarrow P = \sigma^2 (H^T H)^{-1}$$

$$\hat{x} = P H^T \frac{1}{\sigma^2} z = \sigma^2 (H^T H)^{-1} H^T \frac{1}{\sigma^2} z = (H^T H)^{-1} H^T z$$

z₁ led



$$H^T H = \sum t_i^2 \rightarrow \hat{x}_0 = \frac{\sum t_i z_i}{\sum t_i^2}$$

$$H^T z = \sum t_i z_i$$

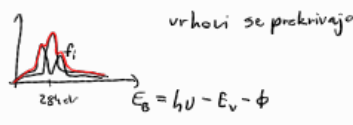
$$\rho = \sigma^2 = \frac{\sigma^2}{\sum t_i^2}$$

$$\rho = \langle (\hat{x}_0 - x_0)^2 \rangle$$

$$H = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \end{bmatrix}$$

Luščenje vrhov v spektrih

• fotoelektr



$$z_i = A f_1(i) + B f_2(i) + r \quad f_1(i) = e^{-(i-\mu)^2 / 2\sigma^2} \text{ Gauss}$$

$$f_2(i) = \frac{\Gamma_0 / 2}{(i-\mu)^2 + \Gamma_0^2} \text{ Lorentz}$$

$$\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad H = \begin{bmatrix} f_1(1) & f_1(2) \\ f_2(1) & f_2(2) \end{bmatrix}, \text{ vse točke naprko } \sigma$$

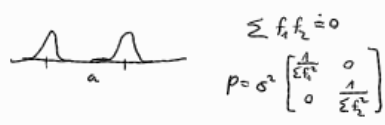
$$R = \sigma^2 I \quad H^T H = \begin{bmatrix} \sum f_1^2 & \sum f_1 f_2 \\ \sum f_1 f_2 & \sum f_2^2 \end{bmatrix}$$

$$\hat{x} = (H^T H)^{-1} H^T z$$

$$\rho = \sigma^2 (H^T H)^{-1}$$

$$(H^T H)^{-1} = \frac{1}{\sum f_1^2 \sum f_2^2 - (\sum f_1 f_2)^2} \begin{bmatrix} \sum f_2^2 & -\sum f_1 f_2 \\ -\sum f_1 f_2 & \sum f_1^2 \end{bmatrix}$$

a) Dobro ločeni vrhovi



b) Deho zlitih vrhovi

$$\sum f_1^2 \sum f_2^2 - (\sum f_1 f_2)^2 \rightarrow 0 \quad f_2 = f_1 + \frac{\partial f_1}{\partial x} a + \dots$$

$$\rho_{12} \neq 0, \quad \rho_{12} \rightarrow -1 \quad f_2^2 = f_1^2 + 2f_1 \frac{\partial f_1}{\partial x} a + \dots$$

$$\rho_{12} = \rho_{21} \sqrt{\rho_{11} \rho_{22}} \quad \sqrt{\sum f_1^2 \sum f_2^2 - (\sum f_1 f_2)^2} = \sqrt{\sum f_1^2 \sum f_1^2 + 2f_1 \frac{\partial f_1}{\partial x} a} =$$

$$\rho_{12} = \frac{-\sum f_1 f_2}{\sqrt{\sum f_1^2} \sqrt{\sum f_2^2}} \quad = \sum f_1^2 \sqrt{1 + \frac{2f_1 \frac{\partial f_1}{\partial x} a}{\sum f_1^2}} =$$

$$= \sum f_1^2 \left(1 + \frac{1}{2} \frac{2f_1 \frac{\partial f_1}{\partial x} a}{\sum f_1^2} \right) =$$

$$= \sum f_1^2 + \sum f_1 \frac{\partial f_1}{\partial x} a$$

$$-\sum f_1 f_2 = -\sum f_1 (f_1 + \frac{\partial f_1}{\partial x} a) =$$

$$= -\sum f_1^2 - f_1 \frac{\partial f_1}{\partial x} a$$

$$\rightarrow \rho_{12} = \frac{-\sum f_1^2 + f_1 \frac{\partial f_1}{\partial x} a}{\sum f_1^2 + \sum f_1 \frac{\partial f_1}{\partial x} a} = -1$$

Lock-in detekcija

• periodično notanje modulatorja v signal

fit funkcija $\sin \omega t$

4tež x_0

$$z = x + r = x_0 \sin(\omega t + \phi) + r$$

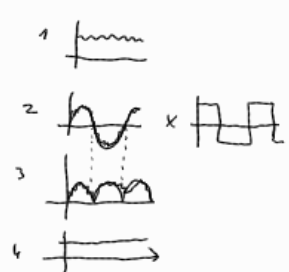
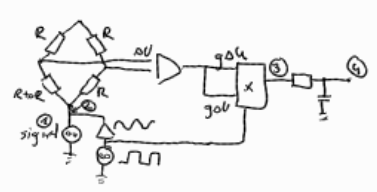
$$H = \begin{bmatrix} \sin \omega t_1 \\ \sin \omega t_2 \\ \vdots \\ \sin \omega t_n \end{bmatrix}$$

$$H^T H = \sum \sin^2(\omega t_i)$$

$$H^T z = \sum \sin \omega t_i z(t_i)$$

$$\hat{x} = \frac{\sum \sin \omega t_i z(t_i) \Delta t}{\sum \sin^2 \omega t_i \Delta t} = \frac{\int_0^T \sin \omega t z(t) dt}{\int_0^T \sin^2 \omega t dt}$$

$$\rho = \frac{\sigma^2}{\sum \sin^2 \omega t_i} = \frac{\sigma^2 \Delta t}{\int \sin^2 \omega t dt} = \frac{2}{T} (\sigma^2 \Delta t) \xrightarrow{T \rightarrow \infty} 0$$



$$z(t) = X_0 \sin(\omega t + \delta)$$

$$\left. \begin{array}{l} \text{REF}_R = \sin \omega t \\ \text{REF}_X = \cos \omega t \end{array} \right\} \begin{array}{l} Y_0 \xrightarrow{\text{LFF}} Y_{0F} \\ Y_{0R} \xrightarrow{\text{LFF}} Y_{0RF} \end{array}$$

$$\left. \begin{array}{l} Y_0 = R \sin \omega t + X_0 \sin(\omega t + \delta) = \frac{1}{2} X_0 R [\cos \delta - \cos(2\omega + \delta)] \rightarrow Y_{0F} = \frac{1}{2} X_0 R \cos \delta \\ Y_{0R} = R \cos \omega t + X_0 \sin(\omega t + \delta) = \frac{1}{2} X_0 R [\sin \delta + \sin(2\omega + \delta)] \rightarrow Y_{0RF} = \frac{1}{2} X_0 R \sin \delta \end{array} \right\} \begin{array}{l} Y_{0F}^2 + Y_{0RF}^2 = \frac{1}{2} X_0 R \\ \tan \delta = \frac{Y_{0RF}}{Y_{0F}} \end{array}$$

Kaj moduliramo?

$$z(t) = X_0 \sin \omega t + r \quad \text{modulirati moramo predel pridežnik,}$$

$$z(t) = X_0 + r \quad \text{drugače po filtriranju še vedno ostane}$$

$V(s) \rightarrow$ odvisno od $B, E, T, \dots$
s modulirano s harmon. sign.

$$s(t) = \bar{s} + A \sin \Omega t$$

$\rightarrow$ LOCK-IN $\rightarrow$ 1. Fourier. komponento pri Ω

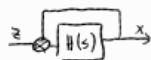
$$V(s(t)) = V(\bar{s}) + \frac{\partial V}{\partial s} \bigg|_{\bar{s}} A \sin \Omega t + \frac{\partial^2 V}{\partial s^2} \bigg|_{\bar{s}} A^2 \sin^2 \Omega t + \dots$$

1. F. komp.

$$\text{out: } \frac{\partial V}{\partial s} \bigg|_{\bar{s}} A$$

- REF: $2s\Omega$ (modulirano nihanje)

$$\text{out: } A^2 \frac{\partial^2 V}{\partial s^2} \bigg|_{\bar{s}}$$



$$\tilde{H} = \frac{H}{1+H}$$

Problem: $H(i\omega) \rightarrow -1$

pri ω , kjer $e^{i\delta} = -1$,

$$|H(i\omega)| = f(\omega) = f=1$$

n-ta iteracija zanke:

$$h=1: -fz$$

$$h=2: -(fz + f^2z)$$

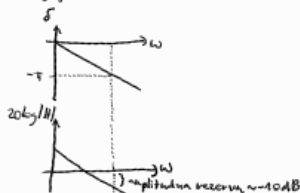
$$h=3: -(fz + f^2z + f^3z)$$

$$N: -\sum_{i=1}^n f^i z$$

$$\rightarrow \sum_{i=1}^n f^i = \frac{1-f^{n+1}}{1-f} \text{ konv., } \text{če } f < 1$$

$$\Rightarrow f < 1 \Rightarrow 20 \log |f| < 0$$

Pogoj za stabilnost



Določitev dušenja: filter 2. reda

$$\tilde{H} = \frac{1 + 2\zeta \frac{s}{\omega_0}}{1 + 2\zeta \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}, \quad \frac{\omega}{\omega_0} = x$$

$$M^2 = |\tilde{H}|^2 = \frac{1 + 4\zeta^2 x^2}{(1-x^2)^2 + 4\zeta^2 x^2} \approx \frac{1 + 2x^2}{1 + x^4}$$

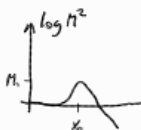
opt. dušenje $\zeta = \frac{1}{\sqrt{2}}$

$$\frac{dM^2}{dx} = 0 = \frac{(1+x^4)(4x) - (1+2x^2)(4x^3)}{(1+x^4)^2} \rightarrow 4x(1+x^4 - x^3(1+2x^2)) = 0$$

$$4x(1-x^2-x^4) = 0$$

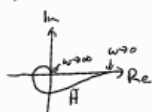
$$x^2 = \frac{-1 \pm \sqrt{1+1}}{2} = \frac{-1 \pm \sqrt{2}}{2} = 0,618$$

$$X_0 = 0,786$$



$$M_0^2 = M^2(x=x_0) = 1,62 \rightarrow M_0 = 1,3 \rightarrow \tilde{H} \leq 1,3$$

Nyquistov diagram $\tilde{H} = \frac{H}{1+H}$



$$H = \text{Re}(H) + i \text{Im}(H) = \xi + i\eta$$

$$M^2 = |\tilde{H}|^2 = \frac{|\xi + i\eta|^2}{|1 + \xi + i\eta|^2} = \frac{\xi^2 + \eta^2}{(1+\xi)^2 + \eta^2}, \quad M \leq M_0$$

$$M^2(1+\xi)^2 + M^2\eta^2 = \xi^2 + \eta^2$$

$$M^2 = \frac{\xi^2(1-M^2) + \eta^2(1-M^2) - 2M^2\xi}{1-M^2}$$

$$\frac{M^2}{1-M^2} = \left(\xi - \frac{M^2}{1-M^2} \right)^2 - \left(\frac{M^2}{1-M^2} \right)^2 + \eta^2$$

$$\frac{M^4 + M^2(1-M^2)}{(1-M^2)^2} = \left(\xi - \frac{M^2}{1-M^2} \right)^2 + \eta^2 = \frac{M^2}{(1-M^2)^2}$$

Premakajeta krošnica

$$\text{radij: } \frac{R}{1-\eta^2}$$

$$\text{center v Re: } \frac{R^2}{1-\eta^2}$$

• krivulja $\subset$ ne sse sekati krožnice

$\Rightarrow$ za $f=1$ mora veljati $-30^\circ < \sigma < 150^\circ$